

MIP Formulations for Delete-Free AI Planning

CPAIOR, 2026

Outline

BACKGROUND AND PRELIMINARY WORK + RESULTS

IMPROVEMENTS TO LM-CUT AND USING IT AS SEPARATOR + RESULTS

Delete-Free Planning : Background & Notation

$$\Pi = \langle P, A, s, t, \text{cost} \rangle$$

- P : Set of boolean variables (facts)
- A : Set of actions, each with
 - $\text{pre}(a) \subseteq P$: Preconditions, facts to be true for the action to be applicable
 - $\text{add}(a) \subseteq P$: Add effects, facts achieved by applying the action
- $s, t \in P$: Initial and Goal facts
- $\text{cost} : A \rightarrow \mathbb{R}_0^+$: Non-negative cost function

Objective: finding the sequence of actions that achieves t , starting from s , that minimizes total cost h^+

Landmarks and LM-Cut

- A disjunctive landmark $L \subseteq A$ is a set of actions such that every plan has at least one action from L
 - Delete-Free planning is equivalent to solving a hitting set problem over all its landmarks
- LM-Cut is a procedure that computes h^{LM} , a relaxation of h^+ :
 - Iteratively: computes a landmark, look up the lowest cost of the actions in such landmark, reduce those action's costs by that amount, repeat
 - h^{LM} is the sum of the costs computed at each iteration

MIP Formulation : Basic model

$$\min \sum_{a \in A} \text{cost}(a) x_a$$

$$\sum_{a \in A_{\rightarrow q}} z_{a,q} = y_q \quad \forall q \in P$$

$$\sum_{a \in A_{p \rightarrow q}} z_{a,q} \leq y_p \quad \forall p, q \in P$$

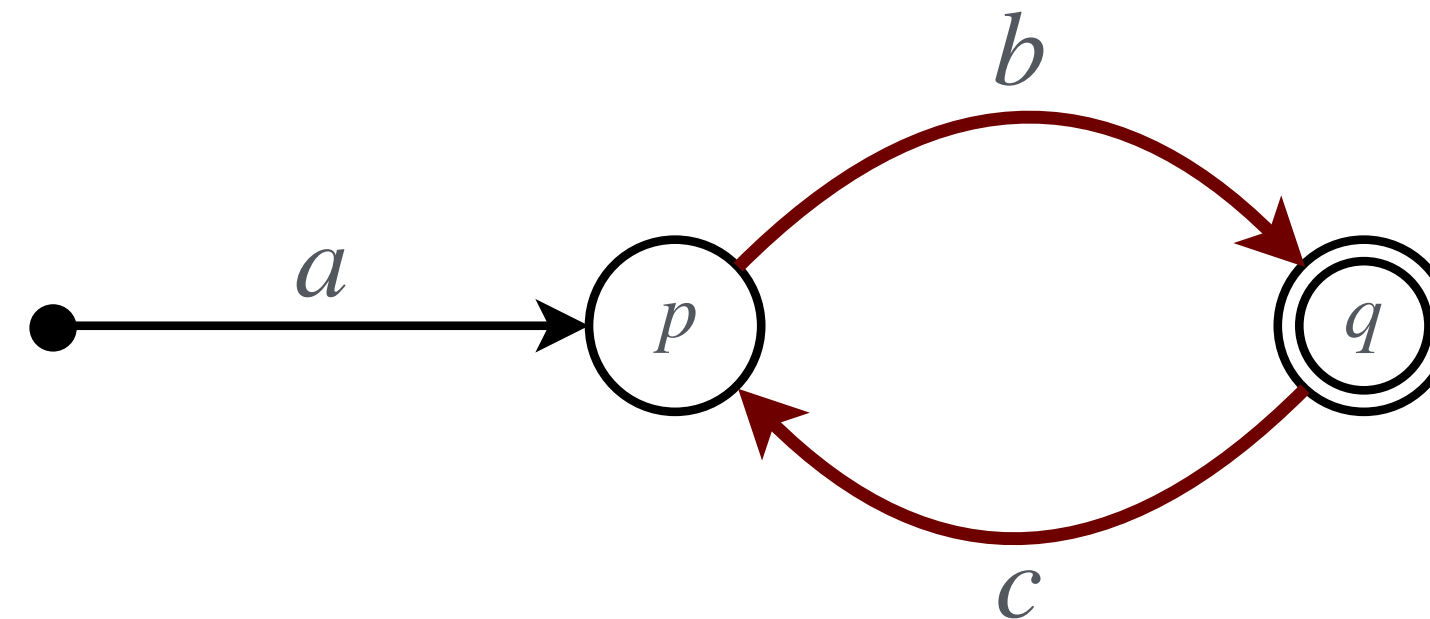
$$z_{a,q} \leq x_a \quad \forall a \in A, \forall q \in \text{add}(a)$$

$$y_s, y_t = 1$$

$$x_a, z_{a,q}, y_p \in \{0,1\} \quad \forall a \in A, \forall q \in \text{add}(a), \forall p \in P$$

MIP Formulation : Acyclicity

- This is not a complete MIP formulation
- We are missing causal acyclicity

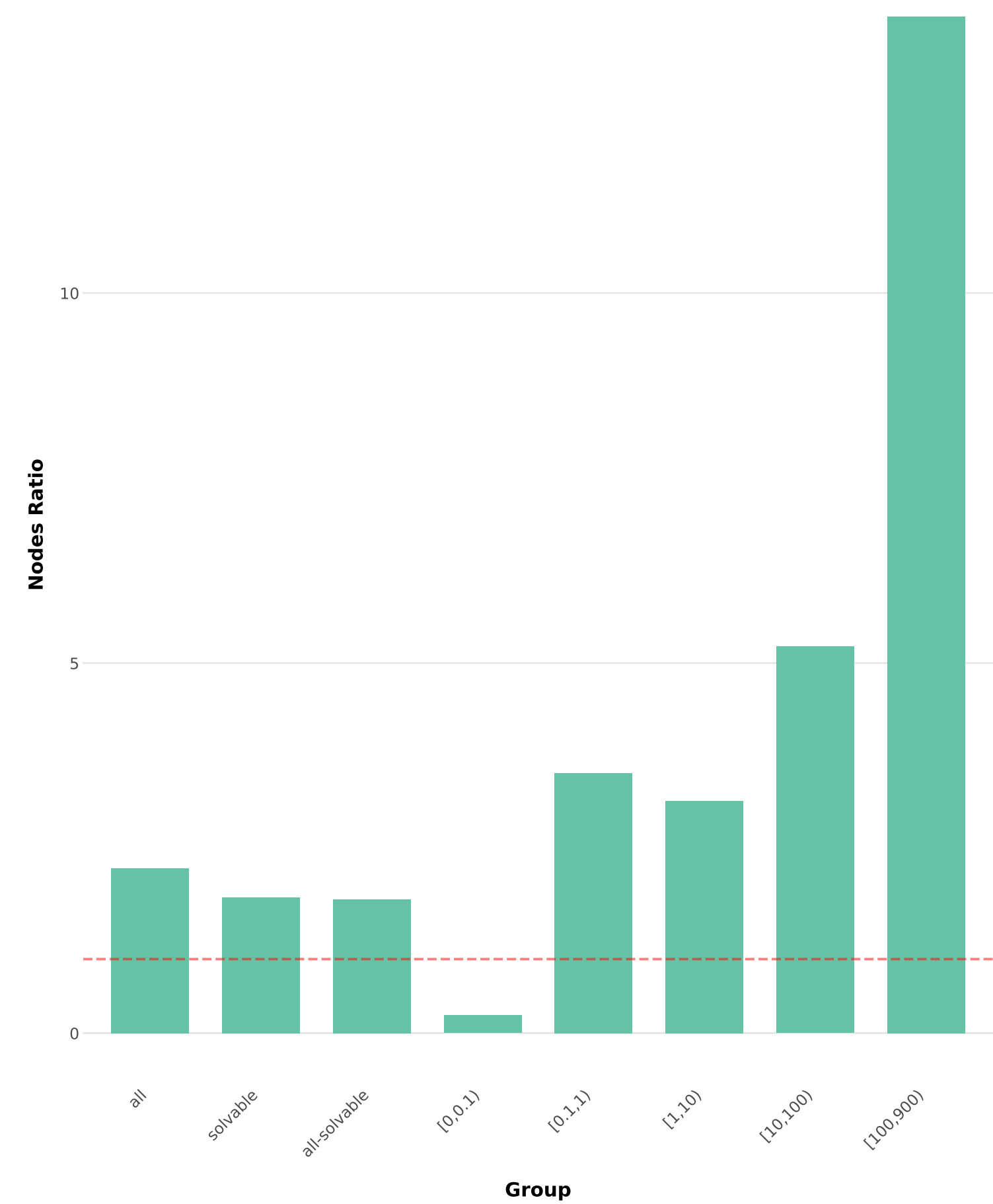
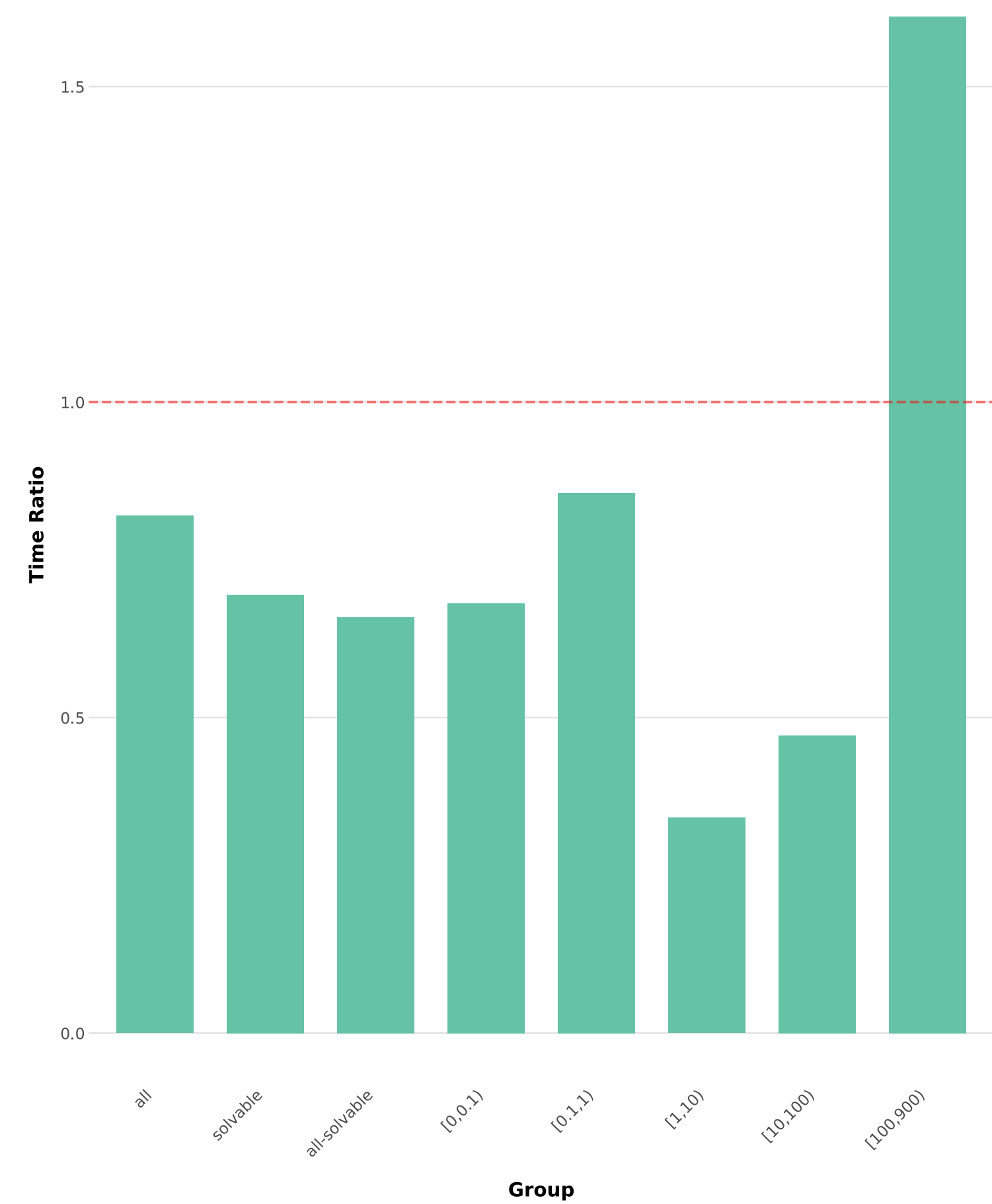


- Vertex Elimination: Previous state of the art used Vertex Elimination graphs to encode precedence (through the transitive closure of the causal relation graph of the task)
- Landmark Constraints: Our approach works by adding to the base model the landmarks violated by its solutions

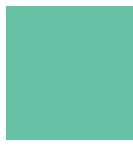
Warm Start

- Primal Heuristic
 - Greedy algorithm
 - Lookahead strategy based on the h^{add} heuristic (a relaxation of h^+) to evaluate different actions
- LM-cut initialization
 - We compute LM-cut and use its landmarks as an initial set of constraints: $\sum_{a \in L} x_a \geq 1$
 - Repeated multiple times to take advantage of its degrees of freedom

Results: current baseline

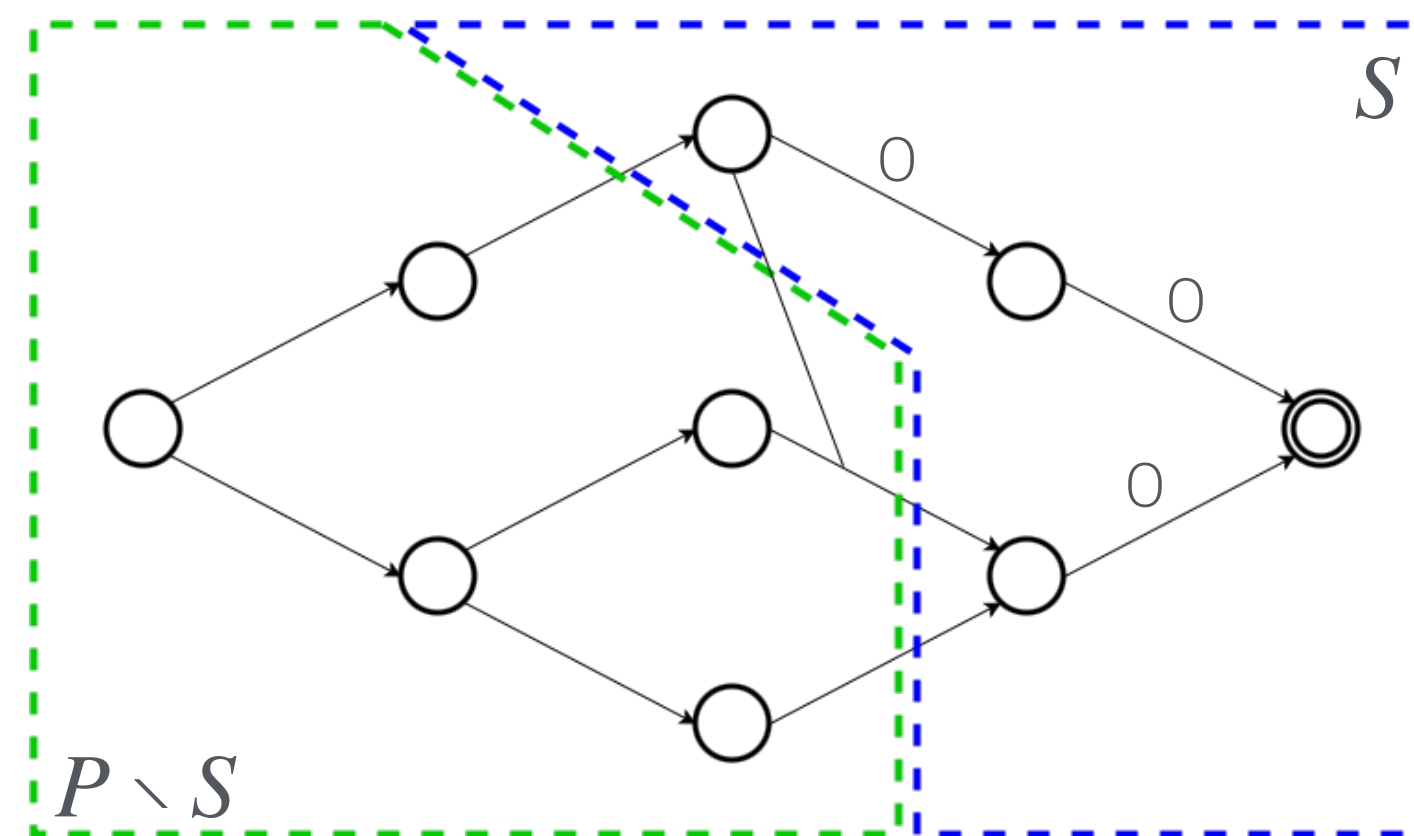


 Vertex Elimination
2496/2799

 Landmark Constraints
2443/2799

Improving LM-Cut

- LM-Cut, at each iteration, builds a justification graph: let $D(a)$ be the precondition with maximum $h^{\max}$ (a further relaxation of h^+) of action a ; the graph is then constructed as $G = (P, E = \{(D(a), q) \mid a \in A, q \in \text{add}(a)\})$ (an edge connecting $D(a)$ to each of a 's effects)
- Partition the facts in goal section S and pre-goal $P \setminus S$, such that S is the set of facts from which t is reachable through a 0-cost path in G

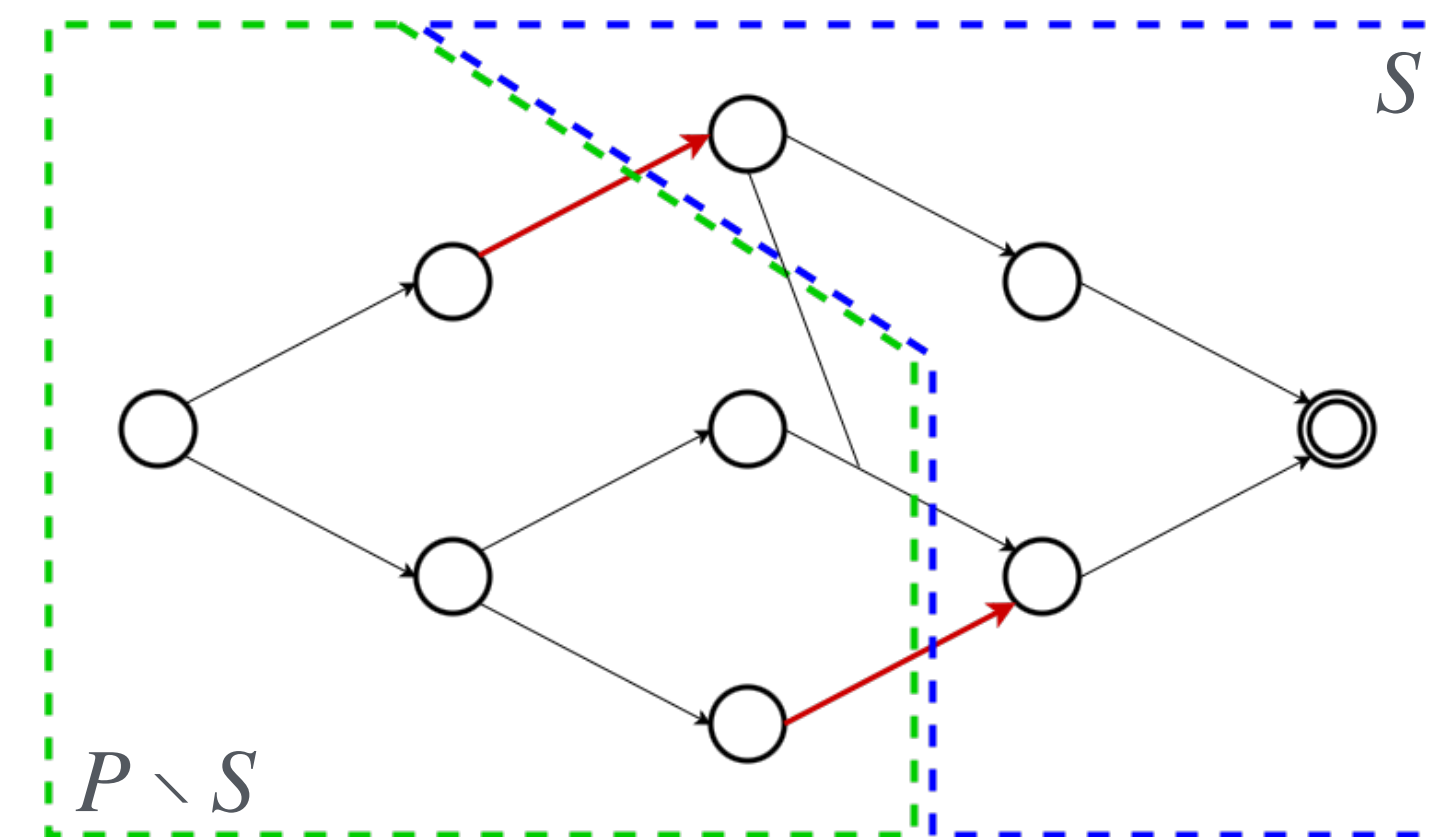
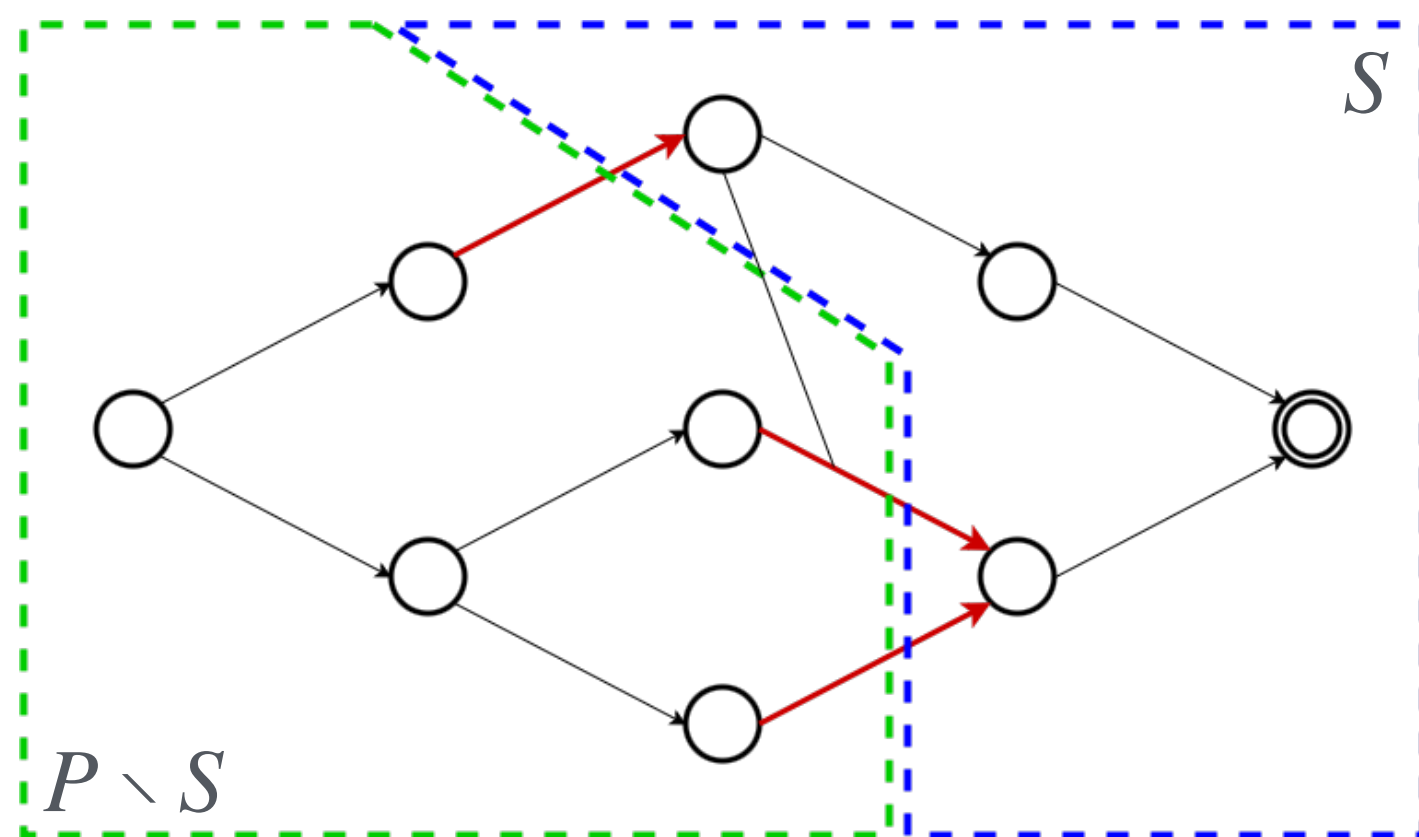


Improving LM-Cut

- The landmark is then computed as the set of actions crossing the partition:

$$L = \{a \in A \mid D(a) \in (P \setminus S), \text{add}(a) \cap S \neq \emptyset\}$$

- L however is not minimal (there might exist a smaller landmark $L' \subset L$):
 - Greedy approach: $L = \{a \in A \mid \text{pre}(a) \subseteq (P \setminus S), \text{add}(a) \cap S \neq \emptyset\}$
 - Extensive approach: remove all actions in L such that it's still is a valid landmark



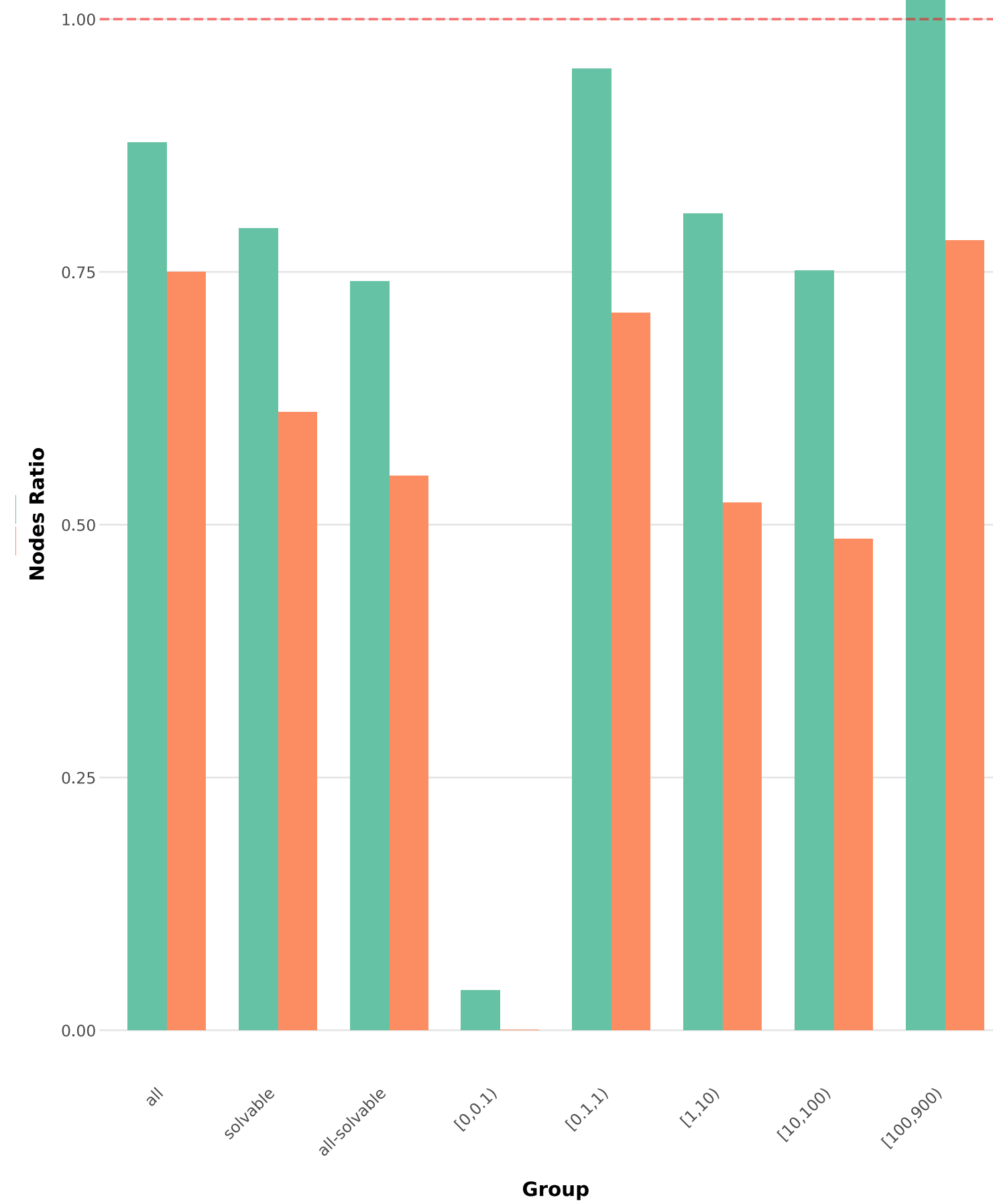
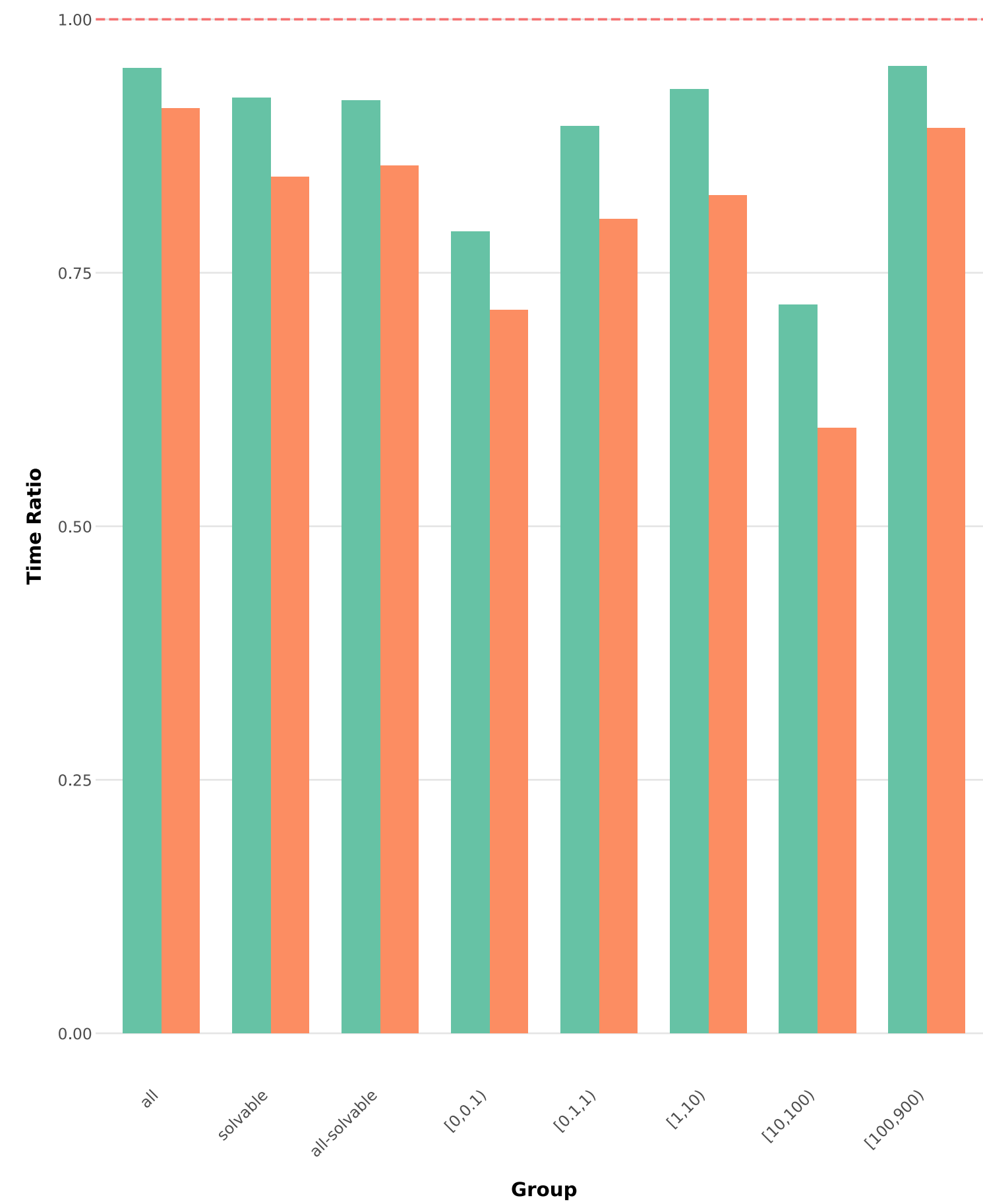
LM-Cut as a cut separator

- By setting the initial cost of a set of actions to 0:
 - The landmarks obtained are still valid, as they are computed as s - t cuts in G
 - Those actions won't appear in any resulting landmark
 - It's as if their cost had been reduced to 0 in "previous iterations"
- We can use LM-Cut as an integer solution separator by setting the initial cost of used actions to 0: all the produced landmarks will then be violating the integer solution as they are all valid landmarks composed only by unused actions

LM-Cut as a cut separator

- Deciding whether a fractional solution violates a landmark is an NP-Hard problem
- Given a fractional solution $\tilde{x}$, we generalize the initial cost assigned to each action:
$$\tilde{c}(a) = (1 - \tilde{x}_a) c(a).$$
- Unlike for integer solutions, there's no guarantee that the landmarks produced will be violated by $\tilde{x}$

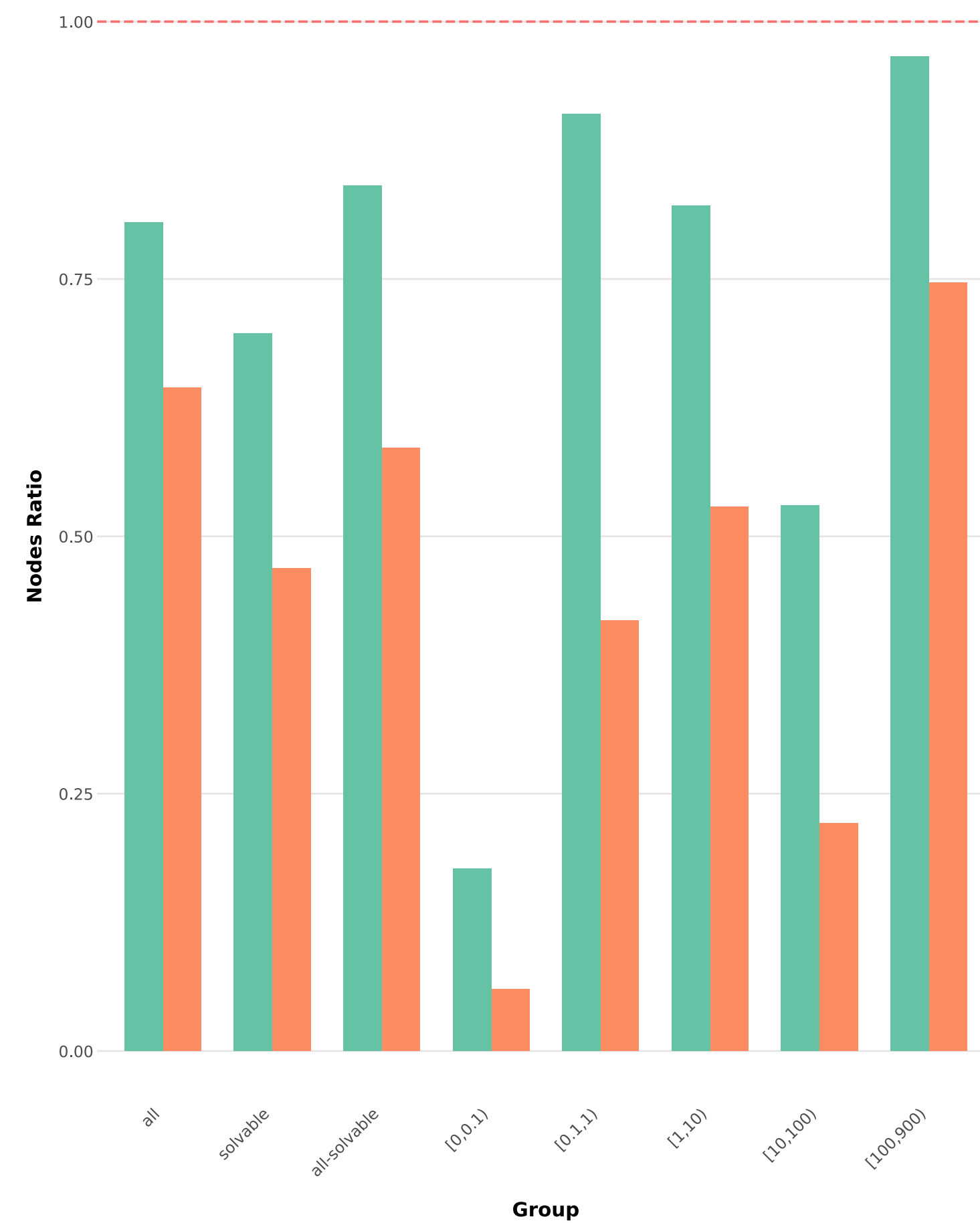
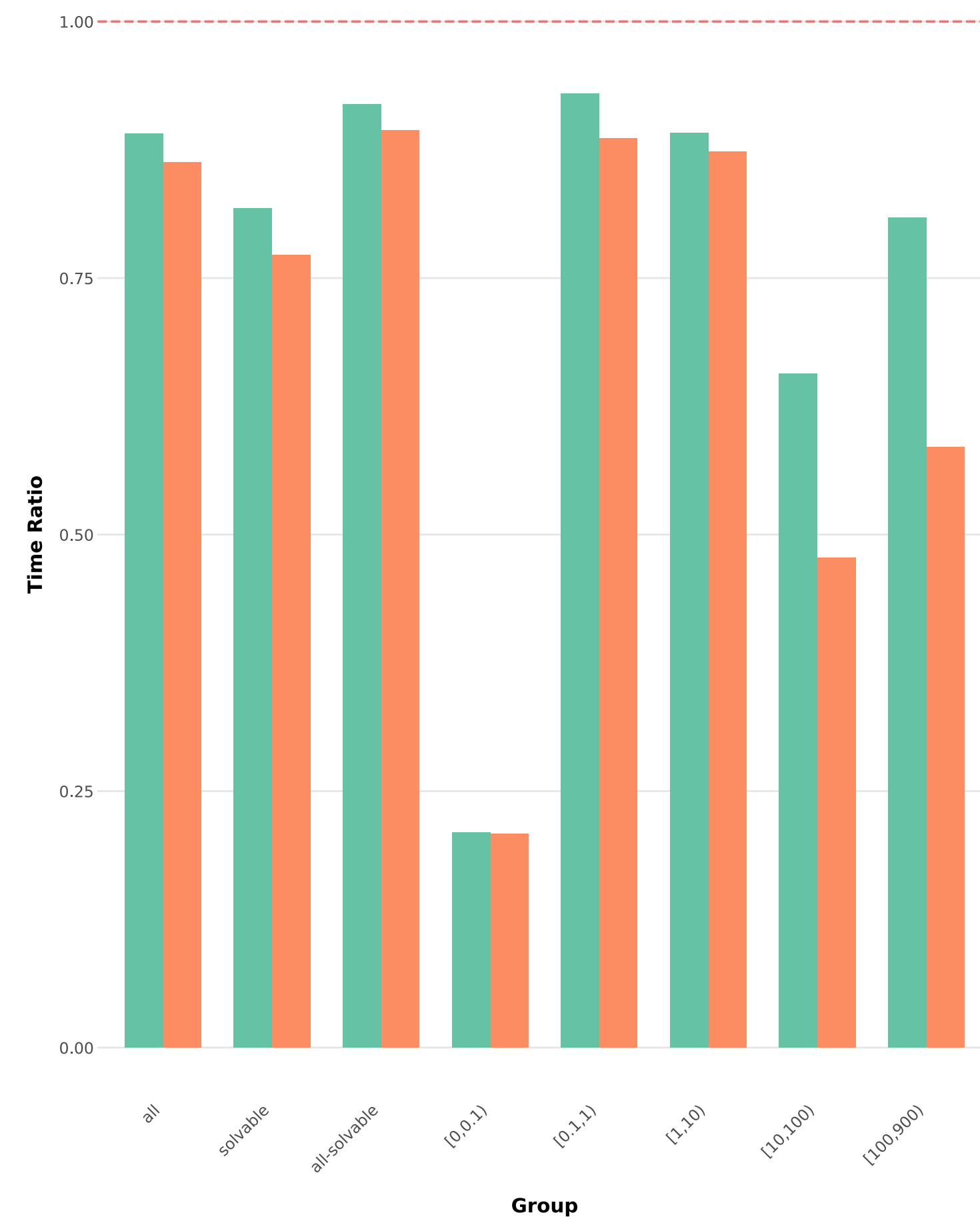
Results: improving LM-Cut



time limit: 15m - memory limit (CPLEX): 4GB

- VE
2496/2799
- VE + LM-Cut improved greedily
2508/2799
- VE + LM-Cut improved extensively
2501/2799

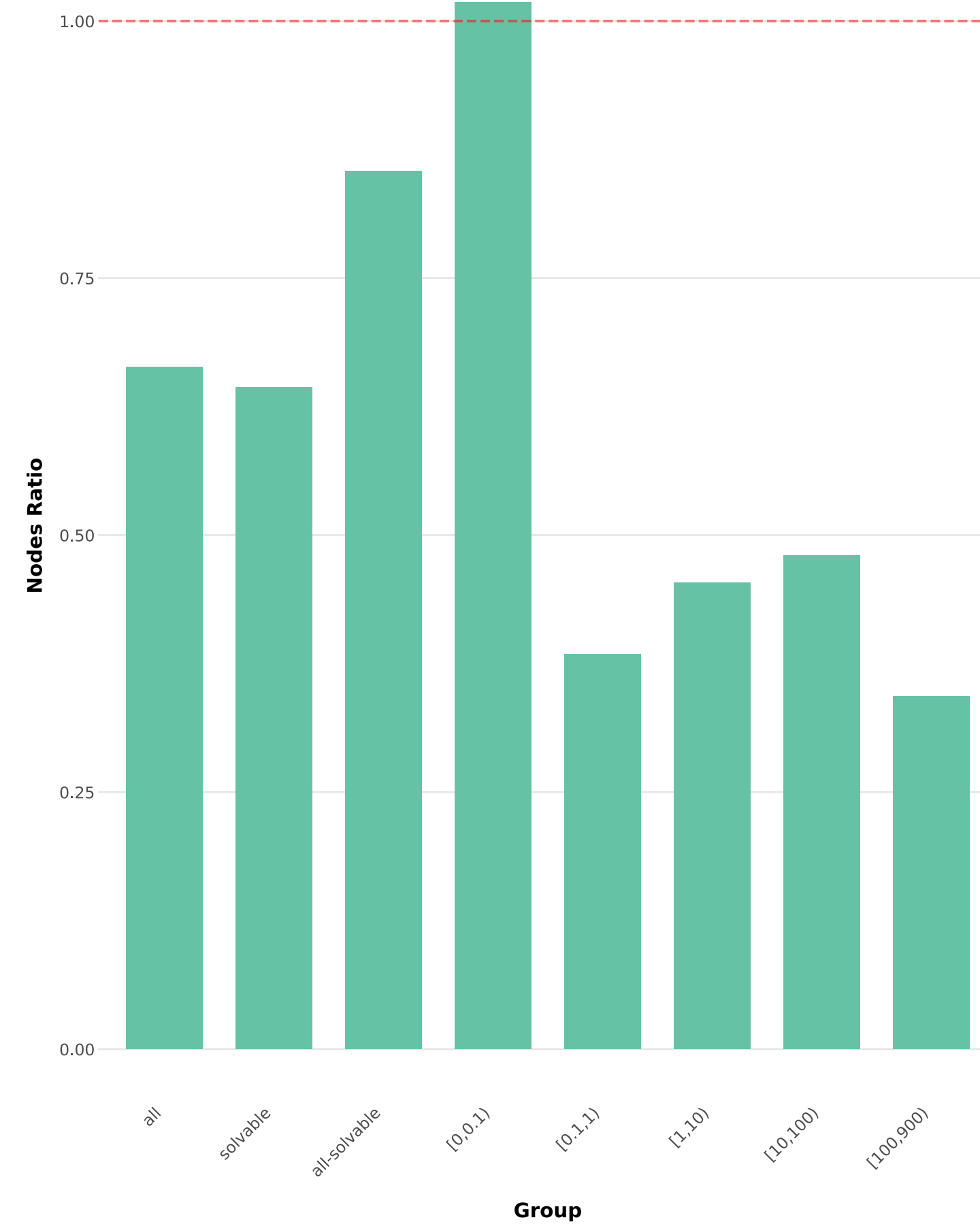
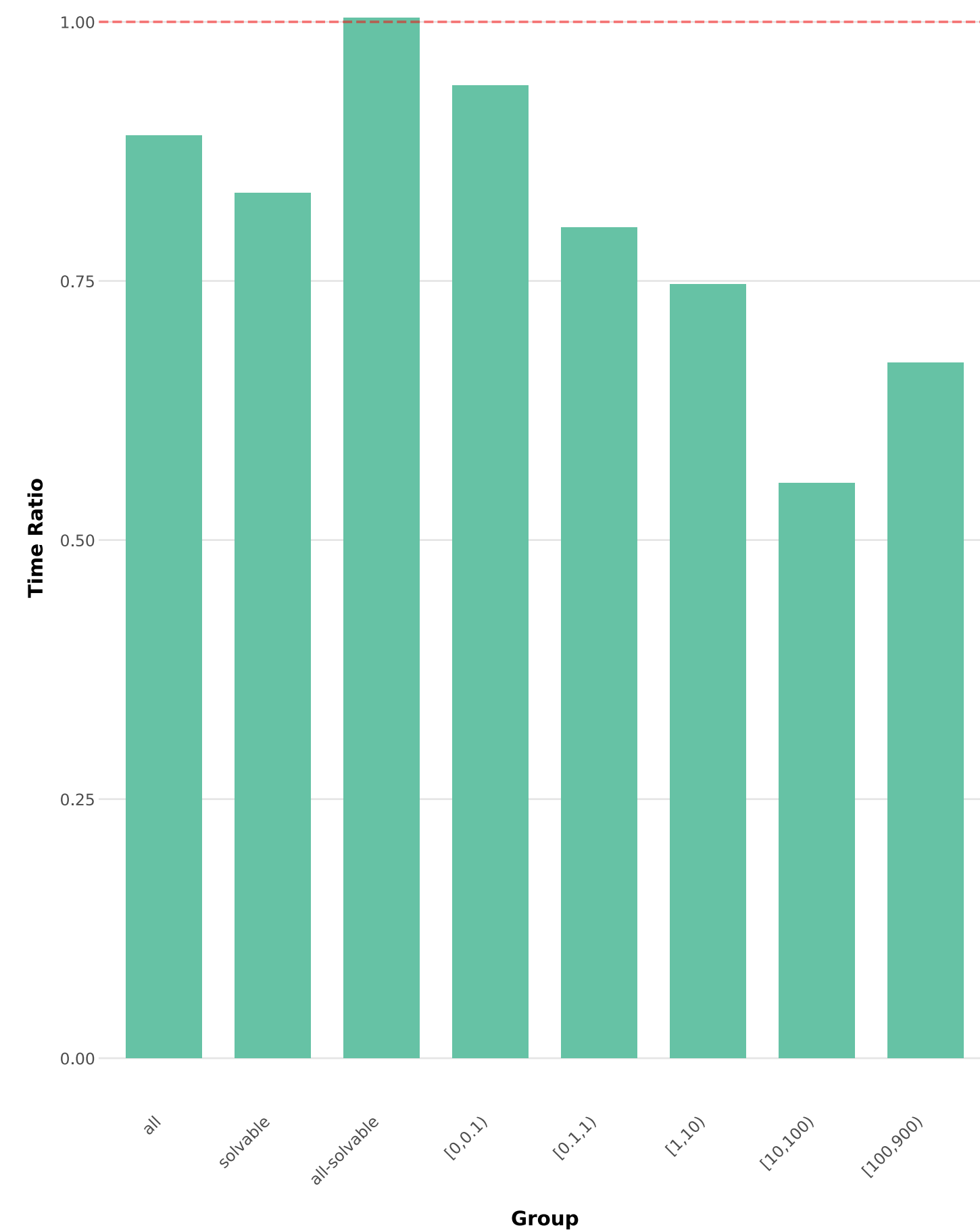
Results: improving LM-Cut



- LM
2443/2799
- LM + LM-Cut improved greedily
2464/2799
- LM + LM-Cut improved extensively
2482/2799

time limit: 15m - memory limit (CPLEX): 4GB

Results: LM-Cut as integer separator

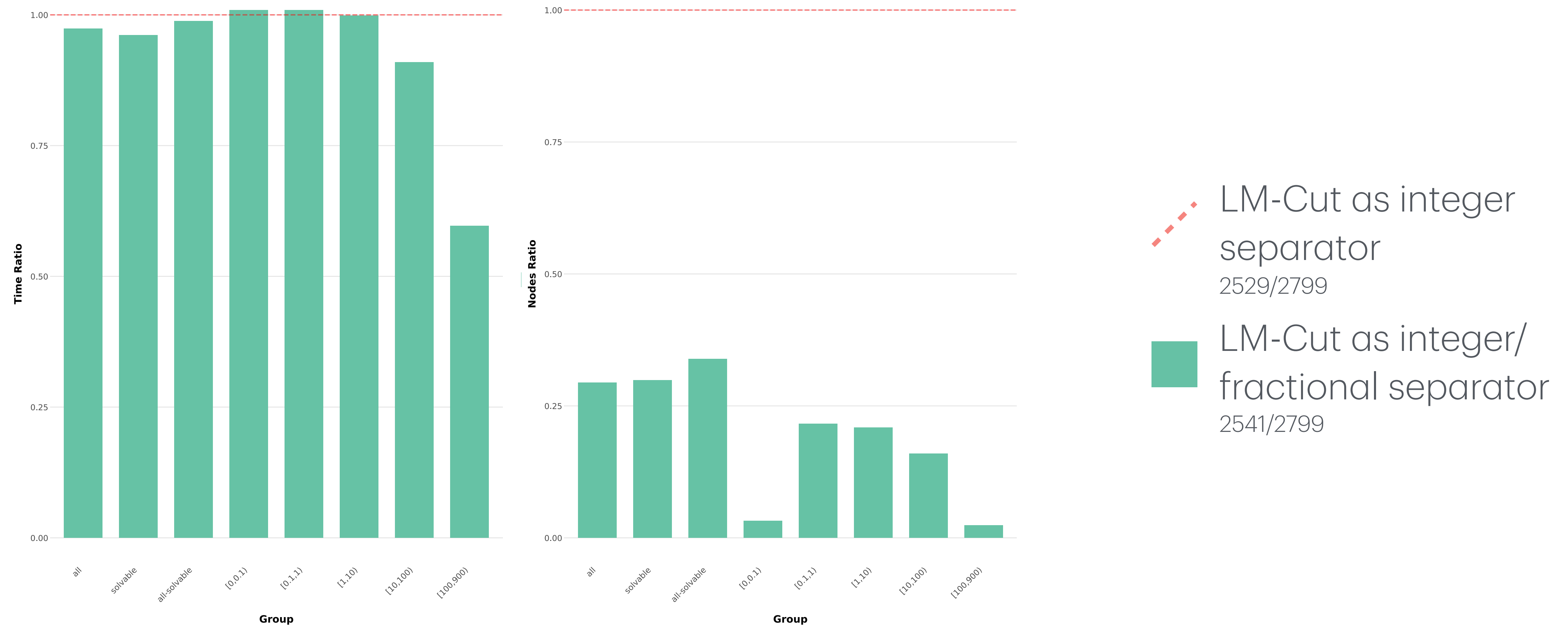


LM (best)
2482/2799

LM-Cut (best) as
integer separator
2529/2799

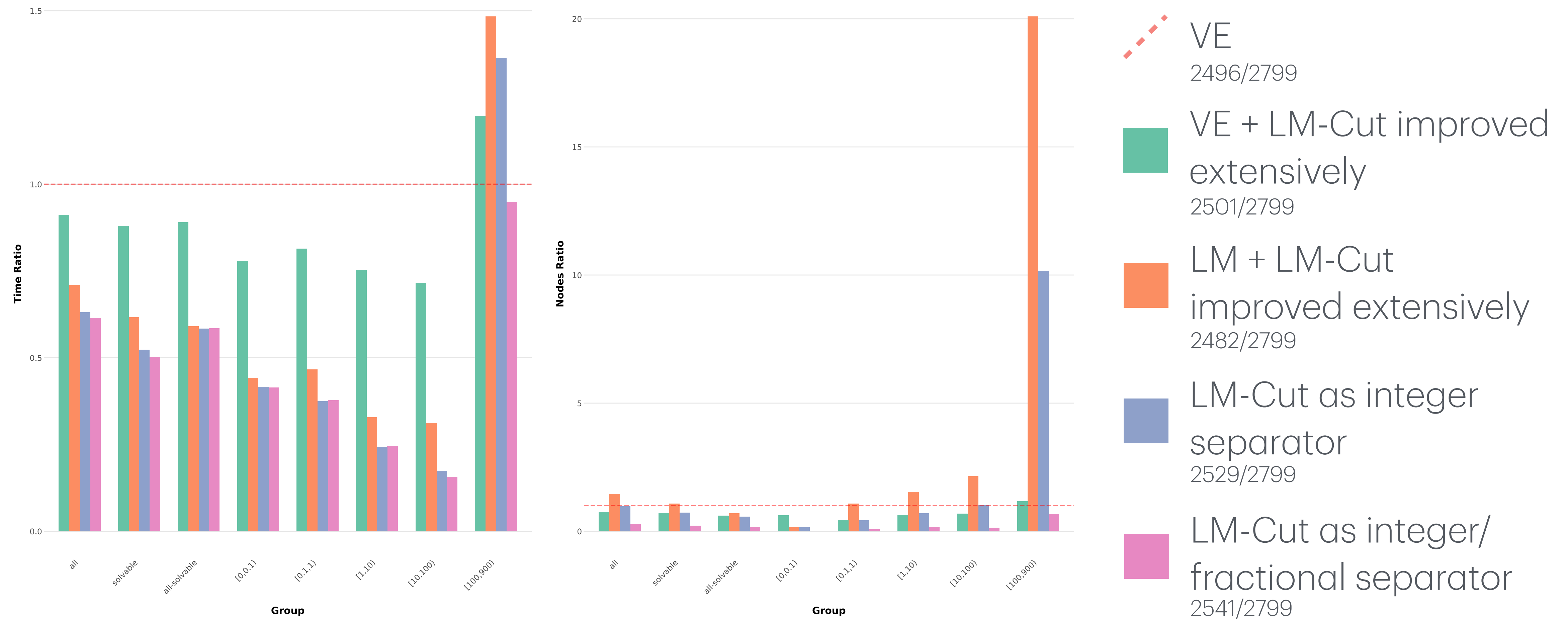
time limit: 15m - memory limit (CPLEX): 4GB

Results: LM-Cut as fractional separator



time limit: 15m - memory limit (CPLEX): 4GB

Results: overall improvements



time limit: 15m - memory limit (CPLEX): 4GB

Conclusions

- Improving LM-Cut to be looking for minimal landmarks provides a ~20% time improvement on both formulations, however, we still have to measure the actual improvement in the h^{LM} bound
- Using LM-Cut to separate landmarks from integer solutions is much more efficient than the previous method, with an additional ~20% time improvement
- Using LM-Cut to separate landmarks from fractional solutions significantly reduces the number of nodes in the B&C tree (~70-80% reduction), with a time improvement for instances in harder time brackets



Thanks for your attention!

If you have any questions, feel free to contact me: matteo.zanella.3@phd.unipd.it

